

Module 01: Systems in High School

Learning Objectives

1. Define **Systems** within the context of High School.
2. Analyze how **Systems** interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of **Systems**.

Introduction

This module explores **Systems**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. **Systems** is a critical component of the 8-part Active Inference spine, bridging the gap between Planning and Agents.

Key Concepts

1. Systems as a Markov Blanket Boundary

How does **Systems** define the boundary between the agent and the environment?

2. Generative Models of Systems

What parameters involved in **Systems** must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of **Systems** drive the perception-action loop?

Applications

In High School, we see **Systems** manifest in: * **Specific Example 1:** A system of linear equations in algebra class is a mathematical Active Inference system: each equation constrains

the possible values of variables (internal states), the solution set represents the system's equilibrium, and adding a new equation (new sensory evidence) further reduces uncertainty -- just as a Markov blanket constrains what an agent can observe about its environment. *

Specific Example 2: A feedback loop in a dynamical system, like the logistic growth equation $P(t+1) = rP(t)(1 - P(t)/K)$, models how a population self-regulates: the current population state generates a prediction of future growth, the carrying capacity K acts as a constraint (Markov blanket), and the system minimizes divergence from equilibrium -- illustrating how mathematical systems naturally minimize free energy.

Conclusion

Understanding Systems allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.